

# On the Hardware-Native Combinatorial Shield: The $R_{yy}$ Local-Cost Isomorphism Extension 9

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## Abstract

The practical utility of barren plateau mitigation strategies depends on their adaptability to native hardware topologies. This note formally investigates Extension 9: the integration of an  $R_{yy}$  interaction basis with an  $m$ -local averaged cost operator. By evaluating the structural independence of the Dynamical Lie Algebra (DLA) and the observable trace combinatorics, we prove that the  $R_{yy}$  architecture strictly preserves the combinatorial variance shield. We establish that this architecture guarantees an amplified asymptotic variance bound of  $\mathcal{R}_m(n) = \frac{n+1}{n+1-m}$  on  $YY$ -coupled systems, proving that topological adaptations do not compromise spatial trace advantages.

## 1 Algebraic Expansion of the $R_{yy}$ Target Space

Let the architecture utilize an  $R_{yy}$  collection block ( $U_{coll}^{(Y)}$ ) yielding generators  $Y_m \otimes Y_i$ , coupled with the standard messenger-controlled CNOT distribution block ( $U_{dist}$ ) yielding generators  $Z_m \otimes X_i$ .

As established in the basis ablation analysis, the non-commutativity of the  $Y_m$  and  $Z_m$  components ensures the generation of the orthogonal  $X_m$  operator space:

$$[Y_m \otimes Y_i, Z_m \otimes X_i] = 2(X_m \otimes Z_i) \quad (1)$$

The iterated Lie brackets strictly close to span the full extended special unitary group. Therefore, the DLA of the  $R_{yy}$  architecture is exactly isomorphic to the  $R_{xx}$  baseline:

$$\mathfrak{g}^{(Y)} = \langle \mathcal{G}_{opr} \cup \mathcal{G}_{coll}^{(Y)} \cup \mathcal{G}_{dist} \rangle_{Lie} \cong \mathfrak{su}(2^{n+1}) \quad (2)$$

Because the algebraic target spaces are identical, both architectures strictly saturate the  $U(2^{n+1})$  Haar measure at depth  $D \rightarrow \infty$ . Consequently, the volumetric denominator in the Master Formula calculation is mathematically invariant under this basis substitution.

## 2 The Invariance of the Trace Combinatorics

The  $m$ -local averaged cost operator  $O_m$  measures strictly the  $n$  data qubits, acting trivially on the messenger. Its traceless part  $\tilde{V}_m$  evaluates to the squared Hilbert-Schmidt trace norm:

$$Tr(\tilde{V}_m^2) = \frac{2N}{\binom{n}{m}} \quad (3)$$

Crucially, the trace operation  $Tr(A^\dagger A)$  is a basis-independent geometric property of the operator space. The substitution of the interaction basis in the unitary circuit driving the state evolution does not alter the geometric norm of the static observable.

### 3 The Asymptotic Variance Bound

We compute the generalized asymptotic gradient variance ratio  $\mathcal{R}_m^{(Y)}(n)$  at Haar saturation by comparing the  $R_{yy}$  Q-LINK trace against a Vanilla circuit operating on  $n + 1$  total qubits with an equivalent  $m$ -local observable.

Because the  $R_{yy}$  DLA saturates the identical Haar measure, the Master Formula constants cancel perfectly. The ratio depends entirely on the invariant combinatorial traces:

$$\mathcal{R}_m^{(Y)}(n) = \frac{\text{Tr}(\tilde{V}_m^2)}{\text{Tr}(V_{\text{Vanilla},m}^2)} = \frac{\frac{2N}{\binom{n}{m}}}{\frac{2N}{\binom{n+1}{m}}} = \frac{\binom{n+1}{m}}{\binom{n}{m}} \quad (4)$$

Expanding the binomial coefficients yields the final geometric bound:

$$\mathcal{R}_m^{(Y)}(n) = \frac{n+1}{n+1-m} \quad (5)$$

**Theorem of Hardware Integration:** The mathematical guarantee of the combinatorial variance shield is independent of the network’s continuous algebraic basis, provided the basis forms a valid isomorphism.

### 4 Conclusion

Extension 9 successfully merges algorithmic theory with hardware reality. The  $R_{yy}$  interaction basis is frequently more native to specific quantum processing units (e.g., certain superconducting cavity architectures) than the  $XX$  interaction. This proof demonstrates that experimentalists can port the Q-LINK architecture to  $YY$ -native hardware and deploy highly non-local cost functions (large  $m$ ) to actively defend against measurement dilution, achieving the exact same mathematically optimal  $\frac{n+1}{n+1-m}$  variance amplification as the theoretical baseline.